

Informe de experimento: results_validation_gpt-5.4_agents

Generado el 2026-09-15 23:37 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-5.4_agents (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f ]
↪ low-agent/benchmarks/results_validation_gpt-5.4_agents
```

- Reports leídos: 1 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: single_call:structured, react_nogate, pagent
- Casos: case14
- Requests: generadas, N=5 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=6, parameterized=3, multistep=3, ambiguous=3
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-15 23:33
- Costo total: 0.3468 USD (15 items; costo medio 0.02312 USD/item)
- Items totales: 15; ok=15; con error=0; tokens totales=117503
- Solved global: 73.3% (11/15)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método single_call:structured

System prompt —1697 caracteres; strategy=structured; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.
```

```
## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

```
## Single-call mode
You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to
↪ call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
↪ and do not fabricate numbers.
```

User prompt (primer item) —3821 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (compact, full case)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
↪ index, transformers use 100000+row index.

Buses

bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
bus 11 | 0.208 kV | load 3.424 MW 1.761 Mvar
bus 12 | 0.208 kV | load 6.54 MW 1.715 Mvar
... (2 líneas omitidas de 14)

Branches

line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
... (8 líneas omitidas de 20)

Available tools

- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Answer by emitting tool calls (function calling) in THIS single response:

- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
↪ non-converged result, say so.

Método react_nogate

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do

You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.

- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format
After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text −67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) −1375 caracteres; architecture=react, gate=False, memory=True

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do
You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format
After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text −67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del **scoreboard_per_case**), en los 4 grupos de **metricas_pfagent_definicion** **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	\$
single_call:structured	case14	5	20.0 (1/5)	4.33e-03	1.58	20.0 (1/5)	100.0 (5/5)	4.68e-03	100.0	−	1.0	4626	−	0.0 (0/5)	0.0693
react_nogate	case14	5	100.0 (5/5)	0	0.00	100.0 (5/5)	100.0 (5/5)	0	100.0	−	2.6	9432	−	0.0 (0/5)	0.1384
pfagent	case14	5	100.0 (5/5)	0	0.00	100.0 (5/5)	100.0 (5/5)	0	100.0	−	2.6	9443	−	0.0 (0/5)	0.1391

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
single_call:structured	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)
react_nogate	100.0 (2/2)	100.0 (1/1)	100.0 (1/1)	100.0 (1/1)
pfagent	100.0 (2/2)	100.0 (1/1)	100.0 (1/1)	100.0 (1/1)

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
single_call:structured	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)
react_nogate	100.0 (2/2)	100.0 (1/1)	100.0 (1/1)	100.0 (1/1)
pfagent	100.0 (2/2)	100.0 (1/1)	100.0 (1/1)	100.0 (1/1)

Verificación final (pfagent: $V(x,c,z,y)$ aplicado a la respuesta final)

pass_first = pasó al primer intento; **pass_retry** = pasó tras el único reintento con el veredicto añadido; **abstained** = falló dos veces, respuesta forzada a fallo declarado sin números. N = items con **final_gate** activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained
pfagent	5	100.0 (5/5)	0.0 (0/5)	0.0 (0/5)

Tipos de error de formulación (conteo de items)

Method	missed_step	ok
single_call:structured	4	1
react_nogate	0	5
pfagent	0	5

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método single_call:structured

Ejemplo: resuelto (solved=True)

```
request (case14-plain-000-s0, case14, difficulty=plain, seed=1813382119): Load case14 and run the power flow, then report the bus with the
↳ lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. Summary: case14 was loaded successfully, but no power-flow result was returned yet, so the lowest-voltage bus cannot be reported
↳ from the available tool outputs.

2. Key numbers: total load 252.6562386075145 MW; generators 5; buses 14; lines 20.

3. Violations: none reported, because no power-flo ...[161 chars más]
--- Respuesta final ---
1. Summary: case14 was loaded successfully, but no power-flow result was returned yet, so the lowest-voltage bus cannot be reported from
↳ the available tool outputs.

2. Key numbers: total load 252.6562386075145 MW; generators 5; buses 14; lines 20.

3. Violations: none reported, because no power-flow or status output was provided.
```

4. Suggested next step: run the power flow and then query network status to identify the bus with the lowest voltage magnitude.

```
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4557;
↪ cost_usd=0.0130; wall=5.2 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
    ↪ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
Loaded case14 successfully.
```

```
2. Key numbers
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 253.84578240591338 MW
- Total generation capacity: 772.4000000000001 MW
```

```
3. Violations, if any
No power-flow or contingency result was returned, and t ...[219 chars más]
--- Respuesta final ---
1. One-sentence summary
Loaded case14 successfully.
```

```
2. Key numbers
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 253.84578240591338 MW
- Total generation capacity: 772.4000000000001 MW
```

```
3. Violations, if any
No power-flow or contingency result was returned, and the requested line disconnection between bus 10 and bus 11 was not executed in the
↪ available tool outputs.
```

```
4. Suggested next step
Run the line disconnection for buses 10-11 and recompute the network status/power flow.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=2.05e-03; F_MAE=1.39; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↪ tokens=4580; cost_usd=0.0135; wall=4.0 s
```

Ejemplo: no resuelto

```
request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↪ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↪ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↪ "to_bus": 13}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,
    ↪ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
I loaded case14 successfully, but the requested line disconnections, rerun, and overloaded-line check were not executed in the available
↪ tool outputs.
```

```
2. Key numbers
- Case: case14
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 257.9016499664 ...[511 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded case14 successfully, but the requested line disconnections, rerun, and overloaded-line check were not executed in the available
↪ tool outputs.
```

```
2. Key numbers
- Case: case14
- Buses: 14
```

- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 257.9016499664761 MW
- Total generation capacity: 772.4000000000001 MW

3. Violations, if any

No post-contingency violations can be reported, because no tool output was provided for:

- disconnecting line 6-13,
- disconnecting line 12-13,
- rerunning the power flow,
- or listing line loadings above 100%.

4. Suggested next step

Execute the remaining tool calls in order: disconnect line 6-13, disconnect line 12-13, rerun/check the solved network status, and then

↪ query branch loadings to list every line ...[11 chars más]

--- Scoring ---

formulation=missed_step (exact=False); V_MAE=1.96e-02; F_MAE=6.49; faithful=100.0 %; solved=False (formulation); flags=-; error=-;

↪ tokens=4736; cost_usd=0.0148; wall=4.8 s

Método react_nogate

Ejemplo: resuelto (solved=True)

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})

executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})

fuelle de la traza: archivo de traza completo

--- Ronda 1 ---

tool: load_case({"case_name": "case14"})

output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,

↪ "total_gen_capacity_mw": 772.4000000000001}

gate: n/a (no es un resultado de flujo)

--- Ronda 2 ---

tool: disconnect_line({"from_bus": 10, "to_bus": 11})

output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,

↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,

↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[5930 chars más]

gate: PASS (reasons=-, mismatch=7.85e-09 MW)

--- Ronda 3 ---

LLM: 1. One-sentence summary

I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the power flow converged successfully.

2. Key numbers

- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.0100 to 1.0900 pu

3. Violations, if ...[443 chars más]

--- Respuesta final ---

1. One-sentence summary

I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the power flow converged successfully.

2. Key numbers

- Total load: 253.85 MW
- Total generation: 266.51 MW
- Total losses: 12.66 MW
- Voltage range: 1.0100 to 1.0900 pu

3. Violations, if any

- Voltage violations only; no thermal violations were reported.
- Overvoltage buses: 1 (1.0600 pu), 6 (1.0700 pu), 7 (1.0573 pu), 8 (1.0900 pu), 11 (1.0636 pu), 12 (1.0536 pu)
- The disconnected line 10-11 has 0.0 MW flow, as expected.

4. Suggested next step

If you want, I can now check the most loaded branch, run an N-1 contingency analysis for this modified network, or suggest remedial actions

↪ for the overvoltage violations.

--- Scoring ---

formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=5311;

↪ cost_usd=0.0167; wall=7.6 s

Ejemplo: primer ítem

request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line

↪ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose

↪ loading is above 100%.

intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,

↪ "to_bus": 13}) -> run_powerflow({})

executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,

↪ "to_bus": 13})

fuelle de la traza: archivo de traza completo

--- Ronda 1 ---

tool: load_case({"case_name": "case14"})

output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,

↪ "total_gen_capacity_mw": 772.4000000000001}

```

    gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 6, "to_bus": 13})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.078737433931405, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000 ...[5832 chars más]
gate: PASS (reasons=, mismatch=4.88e-08 MW)
--- Ronda 3 ---
tool: disconnect_line({"from_bus": 12, "to_bus": 13})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.117618104783695, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01 ...[5729 chars más]
gate: PASS (reasons=, mismatch=5.12e-13 MW)
--- Ronda 4 ---
LLM: 1. Summary: I loaded IEEE case14, disconnected lines 6-13 and 12-13, reran the power flow, and there are no lines loaded above 100%.

2. Key numbers:
- Total load: 257.9016499664761 MW
- Total generation: 273.1440130707419 MW
- Total losses: 15.242363104266296 MW
- Voltage range: 0.9177952034788057 ...[493 chars más]
--- Respuesta final ---
1. Summary: I loaded IEEE case14, disconnected lines 6-13 and 12-13, reran the power flow, and there are no lines loaded above 100%.

2. Key numbers:
- Total load: 257.9016499664761 MW
- Total generation: 273.1440130707419 MW
- Total losses: 15.242363104266296 MW
- Voltage range: 0.9177952034788057 pu to 1.09 pu

3. Violations, if any:
- Lines above 100% loading: none
- Voltage violations are present at:
  - Bus 1: 1.06 pu, overvoltage
  - Bus 6: 1.07 pu, overvoltage
  - Bus 8: 1.09 pu, overvoltage
  - Bus 12: 1.0598406056306475 pu, overvoltage
  - Bus 13: 0.9177952034788057 pu, undervoltage

4. Suggested next step:
- If you want, I can identify the most heavily loaded remaining line or suggest remedial actions to improve the voltage violations,
  ↪ especialmente la low voltage at bus 13.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=11360;
  ↪ cost_usd=0.0326; wall=7.0 s

```

Método pfagent

Ejemplo: resuelto (solved=True)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
  ↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
  ↪ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
  ↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[5930 chars más]
gate: PASS (reasons=, mismatch=7.85e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the post-outage power flow converged.

2. Key numbers
- Total load: 253.8458 MW
- Total generation: 266.5103 MW
- Total losses: 12.6645 MW
- Voltage range: 1.0100 pu to 1.0900 pu

3. Violation ...[470 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the post-outage power flow converged.

2. Key numbers
- Total load: 253.8458 MW
- Total generation: 266.5103 MW
- Total losses: 12.6645 MW
- Voltage range: 1.0100 pu to 1.0900 pu

```

```

3. Violations, if any
- Voltage violations only; no thermal violations were reported.
- Overvoltage buses: 1 (1.0600 pu), 6 (1.0700 pu), 7 (1.0573 pu), 8 (1.0900 pu), 11 (1.0636 pu), 12 (1.0536 pu)
- The disconnected line 10-11 shows 0.0 MW flow, consistent with the outage.

4. Suggested next step
If you want, I can now check the most loaded branch, run an N-1 contingency analysis for this modified network, or recommend remedial
  ↳ actions to reduce the voltage violations.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=5318;
  ↳ cost_usd=0.0168; wall=5.8 s

```

Ejemplo: primer item

```

request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
  ↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
  ↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
  ↳ "to_bus": 13}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
  ↳ "to_bus": 13})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 6, "to_bus": 13})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.078737433931405, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000 ...[5832 chars más]
    gate: PASS (reasons=-, mismatch=4.88e-08 MW)
--- Ronda 3 ---
tool: disconnect_line({"from_bus": 12, "to_bus": 13})
  output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
    ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.117618104783695, "is_violation": false,
    ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": ...[5729 chars más]
    gate: PASS (reasons=-, mismatch=5.12e-13 MW)
--- Ronda 4 ---
LLM: 1. **Summary:** I loaded IEEE case14, disconnected lines 6-13 and 12-13, reran the power flow, and found **no lines loaded above
  ↳ 100%** in the final state.

2. **Key numbers:**
- Total load: **257.9016499664761 MW**
- Total generation: **273.1440130707419 MW**
- Total losses: **15.242363104266 ...[627 chars más]
--- Respuesta final ---
1. **Summary:** I loaded IEEE case14, disconnected lines 6-13 and 12-13, reran the power flow, and found **no lines loaded above 100%** in
  ↳ the final state.

2. **Key numbers:**
- Total load: **257.9016499664761 MW**
- Total generation: **273.1440130707419 MW**
- Total losses: **15.242363104266296 MW**
- Voltage range: **0.9177952034788057 pu to 1.09 pu**

3. **Violations, if any:**
- **Thermal violations:** **None**; there are **no line loadings above 100%**.
- **Voltage violations:**
  - Bus 1: **1.06 pu** overvoltage
  - Bus 6: **1.07 pu** overvoltage
  - Bus 8: **1.09 pu** overvoltage
  - Bus 12: **1.0598406056306475 pu** overvoltage
  - Bus 13: **0.9177952034788057 pu** undervoltage

4. **Suggested next step:**
If you want, I can list the **highest-loaded lines ...[127 chars más]
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=11395;
  ↳ cost_usd=0.0331; wall=7.3 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- $V_MAE \geq 1e-4$ p.u. en 1 filas método×caso con solver: `single_call:structured/case14` (4.33e-03).
- El gate verificó 10 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved más alto: `pfagent` 100.0% (5/5); más bajo: `single_call:structured` 20.0% (1/5).
- Formulación exacta agregada sobre métodos con tools: 73.3% (11/15).
- Formulación por dificultad: mejor `plain` 83.3% (5/6), peor `parameterized` 66.7% (2/3).

- Errores de formulación más frecuentes: `missed_step=4`.
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- 0 de 4 items con mutación de red reportaron estado obsoleto (`stale_state`).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- Costo total 0.3468 USD; el método más caro fue `pfagent` con 0.1391 USD (40 %).
- Métodos con tools: 2.07 tool calls medias por item (máximo 3).

6. Archivos

Rutas absolutas bajo `results_validation_gpt-5.4_agents.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_gpt-5.4_agents/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_agents/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_agents/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_agents/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_agents/scoreboard_per_case.json
traces (15 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_agents/traces
```